

Table of contents

1	Codebook for arcWeeklyTableCombined	2
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1 Codebook for arcWeeklyTableCombined

This codebook describes the variables in the legacy arcWeeklyTableCombined dataset.

Codebook for arcWeeklyTableCombined	...2
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Created by Matthew J. Chandler	NA
NA	NA
General Description	NA

Codebook for arcWeeklyTableCombined	...2
This dataset provides comprehensive information about mobile phone communication events from the first three years of NetSense study (August 2011 through August 2013). Additional participant-reported data on vertex and edge attributes exist in other files; this file contains only observed mobile phone communication data. The unit of observation here is an arc from sender to receiver, where both senders and receivers are either known to be or assumed to be individual persons, identified by unique numeric codes. Any reciprocal arcs (from receiver to sender) are recorded on separate rows, and there are no self-loops. Every arc has at least one study participant, which can be identified by a 5-digit ID code; all other persons have 6-digit ID codes. Most of the data for each arc are recorded in weekly increments—thus detailing the dynamics of communication over time—but the first several variables contain summary information about the arc. Week 1 begins on the official start date of the study on August 15, 2011. Week 0 captures the rare few observations before the official start date. The weekly data are censored at 107 weeks at the latest (ending September 1, 2013), but observation continued through August 2014 (binned in this dataset in week 108+). A minority of arcs are censored earlier: if the participant dropped out of the study early, then the censor date is set for 2/3 the number of total days the participant remained in the study. The censoring allows us to make judgments about whether arcs decayed before the censor date: if the last observation of communication between a sender and receiver occurred before the censor date, then that last observation is treated as the date the arc decayed, but if the last observation occurred after the censor date, then the arc is considered still effectively present as of the censor date. The full list of sender-receiver pairs gives the comprehensive directed network, while data from the weekly increments show the network dynamics. But please note that only mobile phone communication involving study participants is observed, so the largest complete network these data represent includes only the arcs in which both vertices are participants (easily filtered by their 5-digit ID codes).	NA
NA	NA
Variable Name	Vari- able De- scrip- tion

Codebook for arcWeeklyTableCombined		...2
Sender		Unique ID code for the sender of communication (ie, the origin vertex of the arc)
Receiver		Unique ID code for the receiver of communication (ie, the terminal vertex of the arc)
FirstDate		The date of the first observation of communication, given as yyyy-mm-dd

Codebook for arcWeeklyTableCombined		...2
LastDate		The date of the last observation of communication, given as yyyy-mm-dd
DurationDays		The number of days spanning from the first to the last daily observations of communication
FrequencyDays		The total number of days on which at least one communication event was observed

Codebook for arcWeeklyTableCombined	...2
FirstWeekActive	The index number of the week during which the first observation of communication occurred, where week 1 begins on August 15, 2011

Codebook for arcWeeklyTableCombined	...2
LastWeekActive	The index number of the week during which the last observation of communication occurred, where week 1 begins on August 15, 2011
DurationWeeksActive	The number of weeks spanning from the first to the last weekly observations of communication

Codebook for arcWeeklyTableCombined	...2
FrequencyWeeksActive	The total number of weeks during which at least one communication event was observed
FirstWeekEffective	The index number of the week during which the arc is considered to have effectively formed, where week 1 begins on August 15, 2011 (equivalent to First-WeekActive)

Codebook for arcWeeklyTableCombined	...2
LastWeekEffective	The index number of the week during which the arc is considered to have effectively decayed, where week 1 begins on August 15, 2011 (right-censored; see Censor-Week)

Codebook for arcWeeklyTableCombined	...2
DurationWeeksEffective	The number of weeks spanning from First-Week-Effective to Last-Week-Effective (the arc is considered effectively present for all weeks in between)

Codebook for arcWeeklyTableCombined	...2
CensorWeek	The index number of the week upon which the obser- vation of com- munica- tion activity is right- censored (the maxi- mum value possi- ble for Last- Week- Effec- tive)

Codebook for arcWeeklyTableCombined	...2
TotalFrequency__0	The total number of communication events (voice and SMS) observed before August 15, 2011
VoiceFrequency__0	The number of voice communication events observed before August 15, 2011

Codebook for arcWeeklyTableCombined	...2
SMSFrequency_0	The number of SMS communication events observed before August 15, 2011
VoiceLengthSum_0	The sum of the length (in seconds) of voice communication events observed before August 15, 2011

Codebook for arcWeeklyTableCombined	...2
SMSLengthSum_0	The sum of the length (in characters) of SMS communication events observed before August 15, 2011
ActiveDummy_0	A binary indicator of whether any communication events were observed before August 15, 2011

Codebook for arcWeeklyTableCombined	...2
EffectiveDummy_0	A binary indi- cator of whether the arc is con- sidered effec- tively present before August 15, 2011 (equiva- lent to Active- Dummy_0)

Codebook for arcWeeklyTableCombined	...2
ReciprocatedDummy_0	A binary indi- cator of whether recipro- cal commu- nica- tion (from re- ceiver to sender; either voice or SMS) was ob- served before August 15, 2011 ("NA" if never recipro- cated)

Codebook for arcWeeklyTableCombined	...2
ReciprocatedWeight_0	The number of reciprocal communication events (either voice or SMS; counted in 1-to-1 pairs: sender-receiver/receiver-sender) observed before August 15, 2011 (“NA” if never reciprocated)

Codebook for arcWeeklyTableCombined	...2
NonreciprocatedWeight_0	The number of unre- cipro- cated commu- nica- tion events (either voice or SMS; sender- receiver only, net of recipro- cal events) ob- served before August 15, 2011 ("NA" if never recipro- cated)

Codebook for arcWeeklyTableCombined	...2
WeeklyTotalFrequency__w	The total number of communication events (voice and SMS) observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
CumulativeTotalFrequency__w	The cumulative total number of communication events (voice and SMS) observed up to and including week w, where w ranges from 1 to 107
WeeklyVoiceFrequency__w	The number of voice communication events observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
CumulativeVoiceFrequency__w	The cumulative number of voice communication events observed up to and including week w, where w ranges from 1 to 107
WeeklySMSFrequency__w	The number of SMS communication events observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
CumulativeSMSFrequency_w	The cumulative number of SMS communication events observed up to and including week w, where w ranges from 1 to 107
WeeklyVoiceLengthSum_w	The sum of the length (in seconds) of voice communication events observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
CumulativeVoiceLengthSum_w	The cumulative sum of the length (in seconds) of voice communication events observed up to and including week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
WeeklySMSLengthSum_w	The sum of the length (in characters) of SMS communication events observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
CumulativeSMSLengthSum_w	The cumulative sum of the length (in characters) of SMS communication events observed up to and including week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
ActiveDummy__w	A binary indi- cator of whether any commu- nica- tion events were ob- served during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
ActiveAge_w	The age of the active arc as of week w-i.e., the running count of consecutive weeks during which any communication events were observed up to and including week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
ChangeDummy_w	A binary indicator of whether Active-Dummy_w and Active-Dummy_(w-1) are different, where w ranges from 1 to 107
CumulativeChangeFrequency_w	A running count of the number of times Active-Dummy_w changed up to and including week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
EffectiveDummy__w	A binary indi- cator of whether the arc is con- sidered effec- tively present during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
EffectiveAge_w	The age of the effective arc as of week w-i.e., the running count of consecutive weeks during which the arc was considered effectively present up to and including week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
ReciprocatedDummy_w	A binary indi- cator of whether recipro- cal commu- nica- tion (from re- ceiver to sender; either voice or SMS) was ob- served during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
ReciprocatedWeight_w	The number of reciprocal communication events (either voice or SMS; counted in 1-to-1 pairs: sender-receiver/receiver-sender) observed during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
NonreciprocatedWeight_w	The number of unre-ciprocated commu-nica-tion events (either voice or SMS; sender-receiver only, net of reciprocal events) ob-served during week w, where w ranges from 1 to 107

Codebook for arcWeeklyTableCombined	...2
TotalFrequency__108	The total number of communication events (voice and SMS) observed after September 1, 2013
VoiceFrequency__108	The number of voice communication events observed after September 1, 2013

Codebook for arcWeeklyTableCombined	...2
SMSFrequency_108	The number of SMS communication events observed after September 1, 2013
VoiceLengthSum_108	The sum of the length (in seconds) of voice communication events observed after September 1, 2013

Codebook for arcWeeklyTableCombined	...2
SMSLengthSum_108	The sum of the length (in charac- ters) of SMS commu- nica- tion events ob- served after Septem- ber 1, 2013